Finding Surgical Tools with Weak Labels

Project note: classification and localization in robotic surgery video

Lintong Han | lintongbarry.github.io

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1 The problem

In robotic surgery video, knowing which instruments are in view, and where, is a building block for skill assessment, workflow analysis and safety alerts. Labeling every frame by hand is expensive. The robot itself, however, records which instruments are installed on its arms. That gives a cheap but weak label: it says which tools were attached during a clip, not whether each tool is visible in a given frame, and never where it is.

The SurgToolLoc 2022 challenge was built around exactly this setting [1]. Its training set has 24,695 thirty-second clips (60 fps, 720p) from 11 training exercises, each with a single, noisy tool-presence label taken from robot system data and no spatial information. The test set, by contrast, has per-frame bounding boxes drawn by annotators. The task is to classify which of 14 tool types appear and, in a second category, to localize them with boxes.

2 Background: the group and its challenge entry

In 2024 I joined the Vanderbilt group VANDY-VISE as a remote researcher. The group had entered the 2022 challenge before I arrived, and its method is documented in the challenge report [1, Section 3.9]. It used a ResNet backbone with a Query2Label-style transformer decoder [2], where each tool class has its own learned query, trained with a simplified asymmetric loss and a rule that drops the largest per-label loss to cope with noisy labels. Boxes were derived after the fact from the decoder's cross-attention maps: keep the classes predicted present, take the high-energy positions, binarize, and draw bounding boxes around the contours. That entry predates my involvement and is the group's work; it was the starting point for what I did on the same data.

3 What I worked on

Working on the SurgToolLoc 2022 data, I focused on localization, and in particular on segmentation, in this weakly labeled setting.

- **Two kinds of pipeline.** I developed weakly supervised pipelines with a ResNet classifier [3] for which-tool-is-present decisions and a 3D U-Net [4] for where-is-it masks. For video, the third dimension of the 3D U-Net is time: it looks at a short stack of consecutive frames and predicts a mask for the middle frame, so a tool that is briefly occluded in one frame can still be found from its neighbors.
- **Training without spatial labels.** Because training clips carry no masks or boxes, the segmentation model has to learn from pseudo-labels produced by earlier stages, for example from where a classifier finds class evidence or from candidate boxes kept together with their class. Getting these pseudo-labels clean enough is most of the difficulty.
- **Hard frames.** I paid particular attention to frames with occlusion (tools behind tissue or each other) and with several tools at once, where pseudo-labels are least reliable.

- **Benchmarks first.** I set up classification and localization benchmarks so that pipelines could be compared on the same footing, and used them to guide which changes to keep.

4 What I took away

- **A weak label is a statement about the robot, not the image.** "Installed" is not "visible"; a large part of the noise comes from that gap, and methods need to be designed around it.
- **Localization metrics depend on annotation conventions.** Where annotators draw a box on an articulated instrument changes what counts as correct, so it pays to study the reference boxes before tuning a model.
- **Time helps.** Looking at neighboring frames is a cheap way to recover tools that are hidden for a moment.

These lessons carried into later work: in the video restoration and dermatology projects I again started by asking what the labels or references actually measure.

5 Scope of this note

This note describes the problem, the setting and my part at a level I can document. It reports no numbers for my pipelines, and the challenge results in [1] belong to the teams that produced them.

References

[1] A. Zia et al. Intuitive Surgical SurgToolLoc and SurgVU challenges results: 2022–2025. arXiv:2305.07152.

[2] S. Liu, L. Zhang, X. Yang, H. Su, J. Zhu. Query2Label: a simple transformer way to multi-label classification. arXiv:2107.10834, 2021.

[3] K. He, X. Zhang, S. Ren, J. Sun. Deep residual learning for image recognition. *CVPR*, 2016.

[4] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, O. Ronneberger. 3D U-Net: learning dense volumetric segmentation from sparse annotation. *MICCAI*, 2016.